

Learn
Revise
Practice
Excel!

Unit - 3

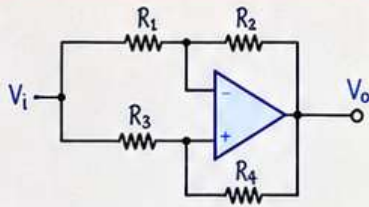
Operational Amplifiers

Complete Handwritten Notes Series (Part 1 - Part 12)

- ✓ Diagrams
- ✓ Formulas
- ✓ Numericals
- ✓ Important Qs
- ✓ Easy to Revise

"Small
Notes
Big
Results!"

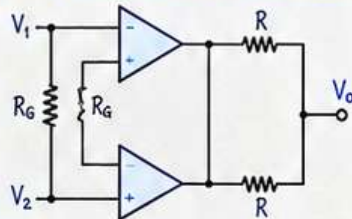
1 Bridge Amplifier



- Circuit Diagram
- Working Principle
- Gain Expression
- Numerical
- Applications

Accurate Differential Amplification

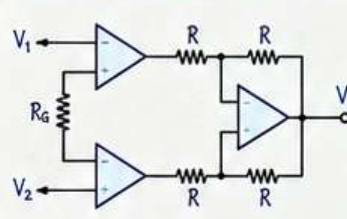
2 2-Op-Amp Instrumentation Amplifier



- Circuit Diagram
- Working
- Gain Derivation
- Numerical
- Applications

High Input Impedance

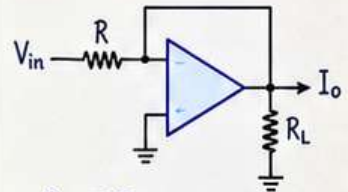
3 3-Op-Amp Instrumentation Amplifier



- Circuit Diagram
- Step-by-step Analysis
- Gain Derivation
- Numerical
- Applications

High CMRR & Accurate Measurement

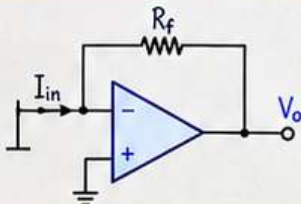
4 V-I Converter



- Circuit Diagram
- Working Principle
- Derivation
- Numerical
- Applications

Convert Voltage to Current

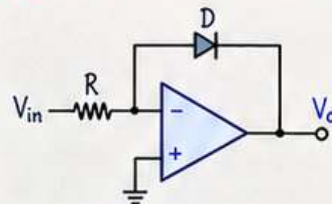
5 I-V Converter



- Circuit Diagram
- Working Principle
- Derivation
- Numerical
- Applications

Convert Current to Voltage

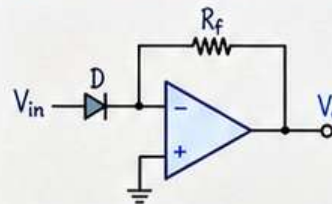
6 Logarithmic Amplifier



- Circuit Diagram
- Working Principle
- Derivation
- Drawbacks
- Applications

$$V_o = -\eta V_T \ln \left(\frac{V_{in}}{V_R} \right)$$

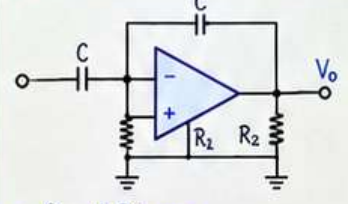
7 Antilogarithmic Amplifier



- Circuit Diagram
- Working Principle
- Derivation
- Applications

$$V_o = -R_f I_s e^{\frac{V_{in}}{\eta V_T}}$$

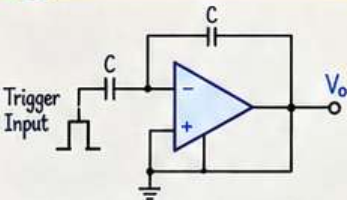
8 Astable Multivibrator



- Circuit Diagram
- Working Principle
- Time Period & Frequency
- Numerical
- Applications

Continuous Square Wave Generator

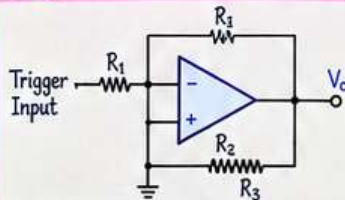
9 Monostable Multivibrator



- Circuit Diagram
- Working Principle
- Pulse Width Derivation
- Numerical
- Applications

Single Pulse of Fixed Width

10 Bistable Multivibrator



- Circuit Diagram
- Working Principle
- Threshold Voltages
- Numerical
- Applications

Two Stable States (Memory Element)

11 Comparison + Important Formula

Parameter	Astable	Monostable	Bistable
No. of stable states	0	1	2
Trigger required	No	Yes	Yes
Output waveform	Square wave	Single pulse	Two-level
Time period	$T = 2RC \ln \left(\frac{1+\beta}{1-\beta} \right)$	$T = RC \ln \left(\frac{1+\beta}{1-\beta} \right)$	—
Main use	Oscillator	Pulse generator	Memory

- All Important Formulas
- Waveforms
- Quick Revision Table

Compare • Revise • Remember

12 Important Questions + Numericals

- ✓ Theory Questions (with answers)
- ✓ Circuit-Based Questions
- ✓ Derivation Questions
- ✓ Numerical Problems (Solved)
- ✓ Expected Exam Questions
- ✓ Short Notes
- ✓ Tips to Solve Numericals



Be Exam Ready!

"Concepts Clear
Doubts Zero"

Study Smart
Score High



Engineering
Builds
Tomorrow!

One Unit
Closer to
Your Goals!



Bridge Amplifier

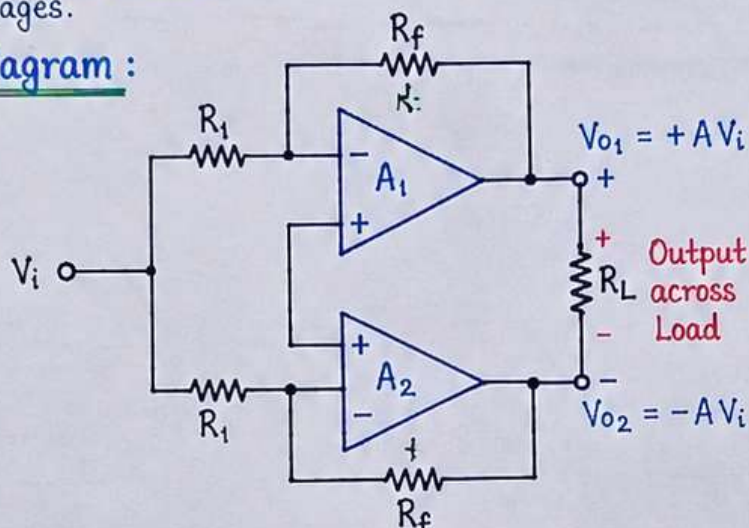
Unit - 3

Special Purpose
Operational Amplifiers

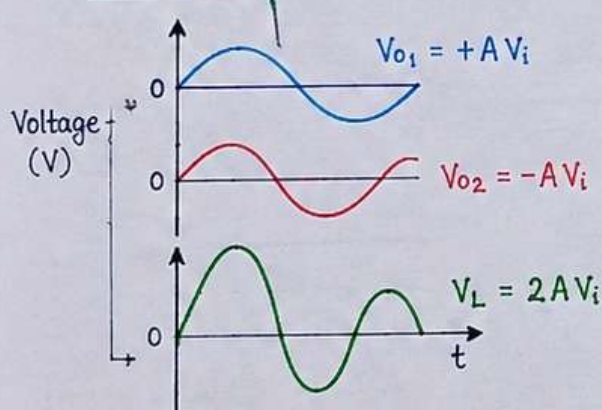
1) Definition :

A bridge amplifier is an amplifier configuration in which two identical amplifiers are connected in a bridge arrangement so that the load is connected between the two output terminals. The output voltage across the load is the difference of the two output voltages.

3) Circuit Diagram :



5) Waveform :



2) Need for Bridge Amplifier :

- To get higher output voltage (double the output).
- To improve power efficiency.
- To reduce even order harmonics and distortion.
- To eliminate the need for large coupling capacitors.
- Useful in audio power amplifiers and speaker driving.

4) Output Voltage :

$$V_{O1} = +A V_i$$

$$V_{O2} = -A V_i$$

Voltage across the load,

$$V_L = V_{O1} - V_{O2} \\ = (+A V_i) - (-A V_i)$$

$$V_L = 2 A V_i$$

∴ Output voltage is doubled.

6) Working :

- ① The same input signal V_i is applied to two identical op-amps.
- ② The first op-amp produces an output in phase with V_i ($V_{O1} = +A V_i$).
- ③ The second op-amp produces an output 180° out of phase ($V_{O2} = -A V_i$).
- ④ The load is connected between the two output terminals.
- ⑤ The load voltage is the difference between the two outputs, which equals $2 A V_i$.
- ⑥ Thus, the effective output voltage across the load is doubled compared to a single amplifier.

7) Advantages :

- Provides higher output voltage (2 times).
- Better power efficiency.
- Reduces even harmonics and distortion.
- No need of large coupling capacitors.
- Useful for low power supply operation.

9) Applications :

- Audio power amplifiers (to drive speakers).
- Telephone line amplifiers.
- Instrumentation and measurement systems.
- Used where high output voltage is required.
- Useful in low voltage, high power applications.

8) Limitations :

- Requires two identical amplifiers.
- Critical matching of components is needed.
- Not suitable for very high frequency applications.
- Circuit is slightly complex.
- Load must be isolated between the two outputs.

10) Key Point to Remember :

- ★ Output across load = Difference of two outputs.
- ★ If gain of each amplifier is A , then $V_L = 2 A V_i$.
- ★ Used to increase output voltage and power.
- ★ Commonly used in audio power amplifier.

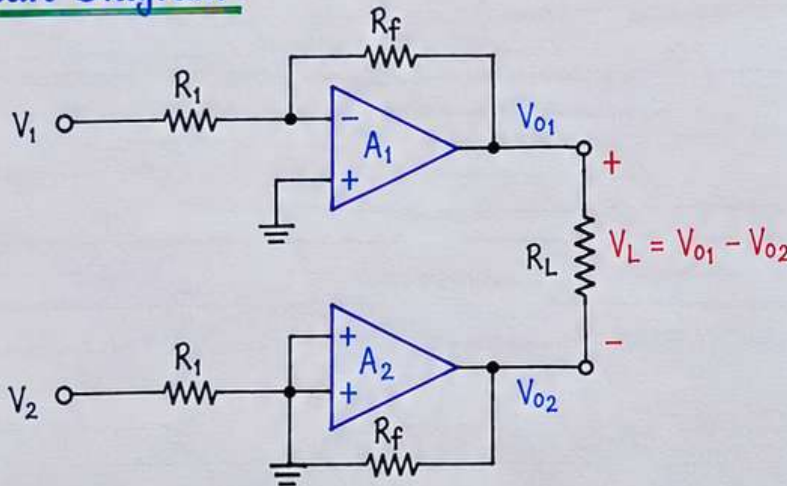
2- Op-Amp Instrumentation Amplifier

Unit - 3
Special Purpose
Operational Amplifiers

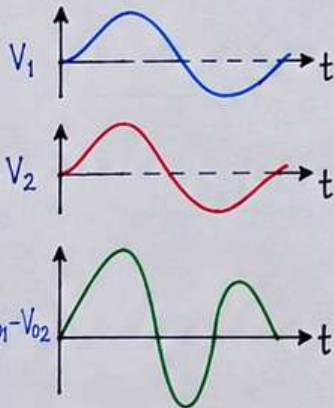
① Definition :

An instrumentation amplifier using two op-amps is a differential amplifier configuration that amplifies the difference between two input voltages while rejecting the common-mode signal. It provides high input impedance and good common-mode rejection, and is mainly used in measurement and instrumentation systems.

③ Circuit Diagram :



⑤ Waveforms :



If V_1 and V_2 are in phase (common signal), ideally $V_L = 0$
If V_1 and V_2 are different, the difference signal is amplified.

② Need / Importance :

- To amplify very small differential signals.
- To reject common-mode noise.
- To provide high input impedance.
- Useful in measurement systems.
- Reduces error due to noise and interference.
- Used in sensor and transducer applications.

④ Output Voltage :

$$V_{01} = -\left(\frac{R_f}{R_1}\right) V_1$$

$$V_{02} = -\left(\frac{R_f}{R_1}\right) V_2$$

Voltage across load,

$$V_L = V_{01} - V_{02} = -\frac{R_f}{R_1} (V_1 - V_2)$$

$$V_L = -\frac{R_f}{R_1} (V_1 - V_2)$$

Here, $V_d = V_1 - V_2$ (Differential Input)

$$\text{Gain, } A_d = \frac{V_L}{V_d} = -\frac{R_f}{R_1}$$

⑥ Working :

- ① Two identical inverting amplifiers are used.
- ② The input voltages V_1 and V_2 are applied to the two op-amps respectively through resistors R_1 .
- ③ Each op-amp produces an output in phase opposition with its input.
- ④ The load is connected between the two output terminals (V_{01} and V_{02}).
- ⑤ The load voltage is the difference between the two outputs, which is proportional to the difference between the two input voltages.
- ⑥ If both inputs are equal (common-mode signal), ideally the output across load is zero.
- ⑦ Thus, it amplifies only the differential signal and rejects the common-mode signal.

⑦ Advantages :

- High input impedance.
- Good common-mode rejection.
- Precise amplification of small signals.
- Reduces noise and interference.
- Simple configuration.
- Useful in measurement and instrumentation.

⑧ Limitations :

- CMRR is lower than 3-op-amp instrumentation amplifier.
- Gain accuracy depends on resistor matching.
- Not suitable for very high gain applications.
- Limited adjustment of gain.

⑨ Applications :

- Sensor signal amplification (e.g. strain gauge, temperature sensor).
- Biomedical instrumentation (ECG, EEG).
- Industrial measurement systems.
- Data acquisition systems.
- Bridge sensor circuits.

10) Key Points to Remember :

- ★ Amplifies the difference between two input signals.
- ★ Rejects common-mode signals.

$$V_L = -\frac{R_f}{R_1} (V_1 - V_2)$$

- ★ Used in measurement and transducer applications.

3- Op-Amp Instrumentation Amplifier

Unit - 3
Special Purpose
Operational Amplifiers

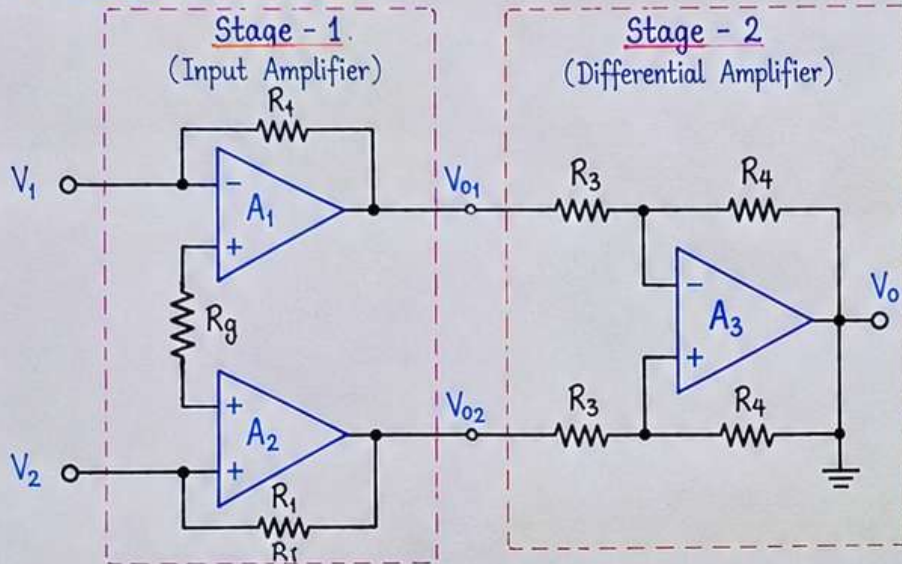
1) Definition :

A 3-op-amp instrumentation amplifier is a high precision differential amplifier that amplifies the difference between two input voltages while rejecting common-mode signals. It consists of two input buffer/amplifier stages and one differential amplifier stage.

2) Need / Importance :

- To amplify very small differential signals.
- To reject common-mode noise (high CMRR).
- To provide high input impedance.
- Used in measurement and instrumentation systems.
- Gain can be easily adjusted.
- Suitable for sensor and transducer applications.

3) Circuit Diagram :



4) Output Voltage / Gain :

Output of first stage :

$$V_{01} = \left(1 + \frac{2R_1}{R_g}\right) V_1$$

$$V_{02} = \left(1 + \frac{2R_1}{R_g}\right) V_2$$

Output of differential amplifier :

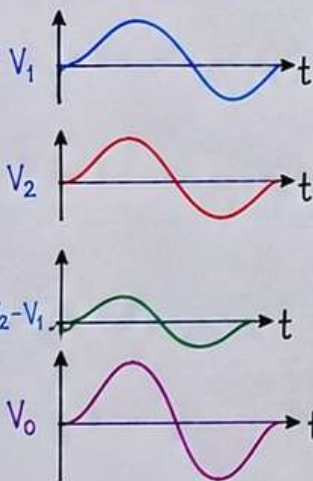
$$V_o = \left(\frac{R_4}{R_3}\right) (V_{02} - V_{01})$$

$$V_o = \left(1 + \frac{2R_1}{R_g}\right) \left(\frac{R_4}{R_3}\right) (V_2 - V_1)$$

Differential Gain,

$$A_d = \frac{V_o}{V_d} = \left(1 + \frac{2R_1}{R_g}\right) \left(\frac{R_4}{R_3}\right)$$

5) Waveforms :



If V_1 and V_2 are in phase (common-mode signal), ideally $V_o = 0$.

If V_1 and V_2 are different, the difference signal is amplified.

6) Working :

- ① The input voltages V_1 and V_2 are applied to the two non-inverting op-amps (A_1 and A_2).
- ② The first stage provides high input impedance and initial gain $\left(1 + \frac{2R_1}{R_g}\right)$.
- ③ The second stage is a differential amplifier that amplifies the difference between V_2 and V_1 .
- ④ If both inputs are equal (common-mode), ideally $V_o = 0$.
- ⑤ The gain can be adjusted by changing R_g or the ratio R_4/R_3 .
- ⑥ The circuit provides high CMRR, high input impedance and accurate differential amplification.

7) Advantages :

- Very high input impedance.
- High common-mode rejection ratio (CMRR).
- Accurate amplification of small signals.
- Gain can be easily adjusted (by R_g).
- Low loading effect on source.
- Suitable for low level signals.
- Widely used in measurement systems.

8) Limitations :

- Requires precise resistor matching (especially R_3 and R_4).
- CMRR depends on component accuracy.
- More components (cost and space).
- Not suitable for very high frequency applications.

9) Applications :

- Strain gauge measurement.
- Temperature sensors (RTD, thermocouple).
- Biomedical instrumentation (ECG, EEG).
- Industrial measurement systems.
- Data acquisition systems.
- Pressure and displacement sensors.

10) Key Points to Remember :

- ★ Three op-amps are used (two input stage + one differential stage).
- ★ Amplifies only the difference between two inputs.
- ★ Rejects common-mode signals.

- ★ Differential gain : $A_d = \left(1 + \frac{2R_1}{R_g}\right) \left(\frac{R_4}{R_3}\right)$
- ★ Used in precision measurement and instrumentation.
- ★ Resistor matching is important for high CMRR.

V-I Converter

(Voltage to Current Converter)

Unit - 3
Special Purpose
Operational Amplifiers

① Definition :

A Voltage to Current (V-I) converter is a circuit which converts an input voltage into a proportional output current. The output current remains constant (independent of load resistance) and is directly proportional to the input voltage.

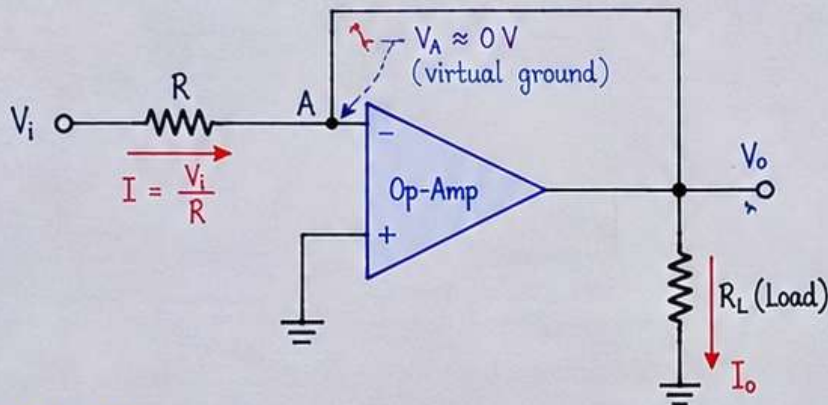
② Need / Applications :

- To drive devices which require constant current (e.g. LEDs, transducers).
- To interface voltage signals with current controlled systems.
- To avoid the effect of load resistance on the output.
- Used in industrial and instrumentation circuits.

③ Basic Concept :

- Op-amp maintains the inverting input at virtual ground ($\approx 0V$).
- The input voltage is converted into a current through an input resistor.
- The same current flows through the load, so the output current is proportional to the input voltage.

④ Circuit Diagram : (Grounded Load)



⑤ Output Current :

At node A, $V_A \approx 0$
Input current, $I = \frac{V_i}{R}$

Since input current of op-amp is ≈ 0 , the same current flows through R_L .

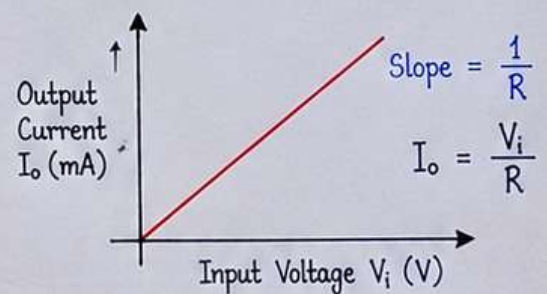
$$I_o = \frac{V_i}{R}$$

So, output current I_o is directly proportional to input voltage V_i and is independent of R_L .

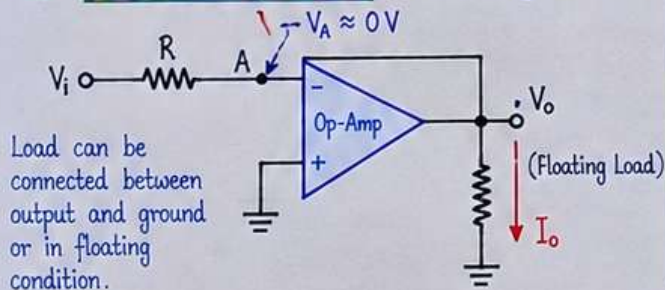
⑥ Working :

- ① The input voltage V_i is applied through resistor R to the inverting input of the op-amp.
- ② Due to negative feedback, the inverting input (node A) is at virtual ground ($V_A \approx 0V$).
- ③ The current through resistor R is $I = \frac{V_i}{R}$.
- ④ Since op-amp input current is nearly zero, the same current flows through the load R_L .
- ⑤ Therefore, the output current $I_o = \frac{V_i}{R}$, which is independent of the load resistance.

⑦ V-I Characteristics :



⑧ Circuit Variation : (Floating Load)



⑨ Advantages :

- Simple and low cost.
- Output current is independent of load resistance.
- Good linearity.
- Wide range of applications.
- High accuracy.
- Can be used to drive current controlled devices.

⑩ Limitations :

- Limited current due to op-amp output drive capability.
- Accuracy depends on resistor tolerance.
- Not suitable for high current applications (requires power op-amp).
- Output compliance voltage is limited.

⑪ Applications :

- Driving LEDs and laser diodes.
- Transducer and sensor circuits.
- Industrial control systems.
- D/A converter (voltage to current).
- 4-20 mA current loop systems.
- Current source for measurement circuits.
- Battery charger and current regulator circuits.

⑫ Example :

If $V_i = 2V$ and $R = 1k\Omega$, find the output current I_o .

Solution :

$$I_o = \frac{V_i}{R} = \frac{2}{1000}$$

$$I_o = 2 \text{ mA}$$

⑬ Key Points to Remember :

- ★ $I_o = \frac{V_i}{R}$
- ★ Output current is independent of load resistance.
- ★ Inverting input is at virtual ground.
- ★ Used to convert voltage signal into current signal.
- ★ Commonly used in LED driving and instrumentation.

I-V Converter

(Current to Voltage Converter)

Unit - 3
Special Purpose
Operational Amplifiers

① Definition :

An I-V converter (Current to Voltage Converter) is a circuit which converts an input current into a proportional output voltage. It is also called Transimpedance Amplifier. The output voltage is directly proportional to the input current and is independent of the load resistance (for ideal op-amp).

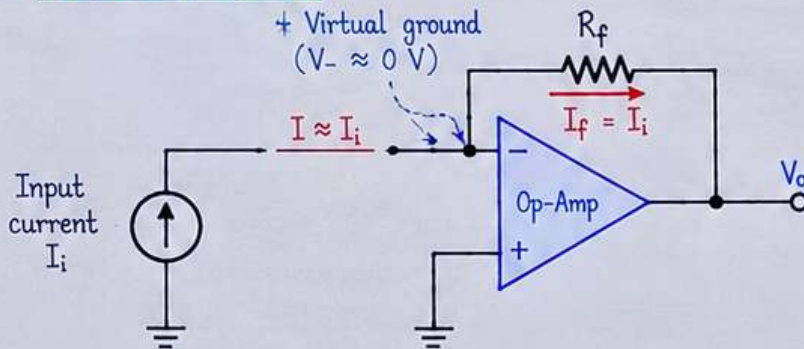
② Need / Applications :

- To convert current output of sensors (e.g. photodiode) into a measurable voltage.
- To measure very small currents.
- To interface current sources with voltage measurement systems (ADC).
- Used in optical, biomedical and industrial instrumentation.
- Provides high sensitivity for low current signals.

③ Basic Concept :

- Op-amp maintains the inverting input at virtual ground ($\approx 0V$).
- The entire input current flows through the feedback resistor R_f .
- The output voltage develops across R_f .
- Output voltage is proportional to input current.
- The load resistance does not affect the output voltage (ideally).

④ Circuit Diagram :



⑤ Output Voltage / Formula :

At inverting input (virtual ground), $V_- \approx 0V$
By KCL at node V_- ,
 $I_i = I_f$

$$I_f = \frac{V_o}{R_f}$$

$$V_o = -I_i R_f$$

Thus, output voltage is proportional to input current and negative in sign (phase inversion).

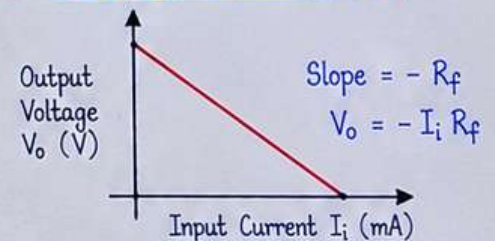
Transimpedance (Gain),

$$A_v = \frac{V_o}{I_i} = -R_f \text{ (in } \Omega \text{)}$$

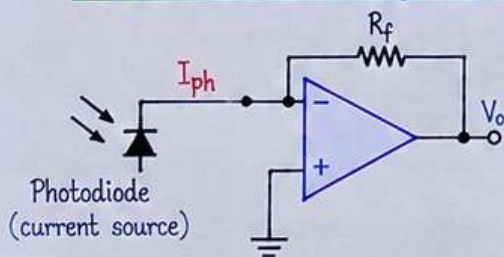
⑥ Working :

- ① The input current I_i is applied to the inverting input of the op-amp.
- ② Due to negative feedback, the inverting input remains at virtual ground ($V_- \approx 0V$).
- ③ The entire input current flows through the feedback resistor R_f .
- ④ A voltage drop is developed across R_f , producing an output voltage $V_o = -I_i R_f$.
- ⑤ The output voltage is independent of the load resistance (ideally).
- ⑥ If the direction of input current is reversed, the polarity of output voltage also reverses.

⑦ I-V Characteristics :



⑧ Practical Circuit using Photodiode :



Used to convert small photocurrent into a measurable voltage.

⑨ Advantages :

- High sensitivity (can measure very small currents).
- Simple configuration.
- Output independent of load resistance.
- Wide range of applications.
- Good linearity.
- Accurate and stable output.

⑩ Limitations :

- Accuracy depends on feedback resistor tolerance.
- Limited bandwidth (R_f and input capacitance affect speed).
- Not suitable for very high current signals.
- Offset voltage and bias current cause error in measuring small currents.

⑪ Applications :

- Photodiode based light sensors.
- Radiation detectors (X-ray, γ -ray).
- Biomedical instruments (ECG, EEG).
- Industrial current measurement.
- Chemical and gas sensors.
- Optical communication systems.
- Data acquisition and measurement systems.

⑫ Example :

A current of $100 \mu A$ is applied to an I-V converter with $R_f = 10 k\Omega$. Find the output voltage.

Solution :

$$V_o = -I_i R_f = -(100 \times 10^{-6}) \times (10 \times 10^3)$$

$$V_o = -1V$$

⑬ Key Points to Remember :

- ★ $V_o = -I_i R_f$
- ★ Input applied at inverting input.
- ★ Inverting input is at virtual ground.
- ★ Output independent of load resistance.
- ★ Also called Transimpedance Amplifier.
- ★ Widely used with photodiode.
- ★ Used to convert current signal into voltage signal.

Logarithmic Amplifier

Unit - 3
Special Purpose
Operational Amplifiers

① Definition :

A logarithmic amplifier is an op-amp circuit whose output voltage is proportional to the logarithm of the input voltage. It uses a diode or transistor in the feedback path, which has an exponential V-I characteristic. Thus, it performs the mathematical operation of taking the logarithm of the input signal.

② Need / Applications :

- To compress a wide range of input voltages into a smaller output range (log scale).
- Useful in measurement of signal levels in dB.
- Used in analog computation (multiplication, division).
- Used in automatic gain control (AGC).
- Useful in signal processing, telecommunication and audio systems.

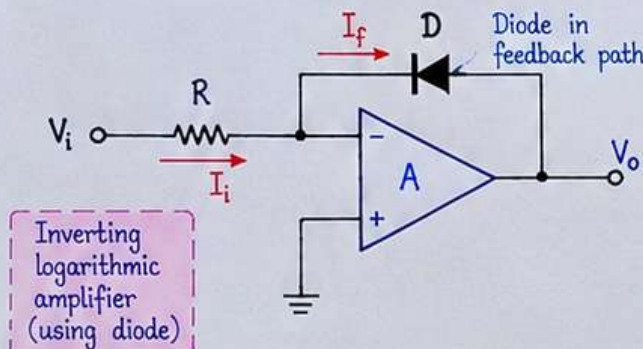
③ Basic Concept :

- A p-n junction diode/transistor has an exponential current-voltage relationship :

$$I_D = I_S \left(e^{\frac{V_D}{\eta V_T}} - 1 \right)$$

- For $V_D \gg \eta V_T$, $I_D \approx I_S e^{V_D/\eta V_T}$
- In a logarithmic amplifier, this exponential relation is used in feedback to obtain an output voltage proportional to log of the input voltage.

④ Circuit Diagram (Using Diode) :



⑤ Derivation :

- Current through input resistor :

$$I_i = \frac{V_i}{R}$$

- Since input current of op-amp is ≈ 0 , $I_i = I_f = I_D$

- For diode, $I_D = I_S e^{V_D/\eta V_T}$

- At inverting input, $V_D = -V_o$ (since non-inverting input is at 0 V).

- So, $I_i = I_S e^{-V_o/\eta V_T}$

$$\frac{V_i}{R} = I_S e^{-V_o/\eta V_T}$$

- Taking log (both sides),

$$V_o = -\eta V_T \ln \left(\frac{V_i}{R I_S} \right)$$

Where,

I_S = Reverse saturation current of diode

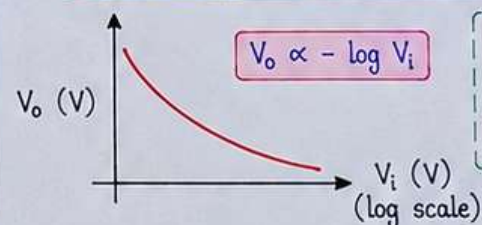
V_T = Thermal voltage (≈ 26 mV at room temperature)

η = Ideality factor (≈ 1 to 2)

⑥ Working :

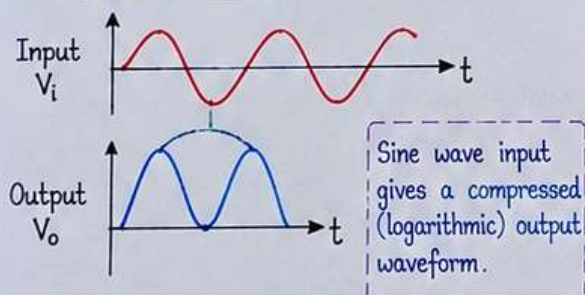
- ① The input voltage V_i is applied through resistor R to the inverting input of the op-amp.
- ② Due to virtual ground, the inverting input is at 0 V ($V \approx 0$).
- ③ The current through R is $I_i = \frac{V_i}{R}$, which flows through the diode in the feedback path.
- ⑤ The diode has an exponential I-V characteristic.
- ⑦ As input voltage increases, the diode current increases exponentially.
- ⑥ The op-amp output adjusts to keep the inverting node at virtual ground, resulting in an output voltage proportional to the logarithm of the input.

⑦ Input-Output Characteristic :



As V_i increases (log scale), V_o decreases linearly.

⑧ Waveforms :



⑨ Advantages :

- Can handle wide range of input voltages.
- Useful for log conversion and dB measurements.
- Simple circuit.
- Good accuracy for low frequency signals.
- Used in analog signal processing.

⑩ Limitations :

- Accuracy depends on diode characteristics (I_S , η).
- Temperature sensitive (V_T changes).
- Limited input voltage range.
- Not suitable for high precision applications.
- Non-linearities at very low input levels.

⑪ Applications :

- Measurement of signal strength in dB (e.g. RF, audio systems).
- Analog multiplication and division circuits.
- Automatic gain control (AGC).
- Audio and video signal processing.
- Compression of dynamic range.
- Used in communication systems and instrumentation.

⑫ Key Points to Remember :

- ★ Uses diode/transistor in feedback path.
- ★ Output is proportional to log of input voltage.
- ★ $V_o = -\eta V_T \ln \left(\frac{V_i}{R I_S} \right)$.
- ★ Provides compression of wide input range.
- ★ Commonly used for dB measurement.

Antilogarithmic Amplifier

Unit - 3
Special Purpose
Operational Amplifiers

① Definition :

An antilogarithmic amplifier is an op-amp circuit whose output voltage is proportional to the antilogarithm (exponential) of the input voltage. It performs the inverse operation of a logarithmic amplifier. It uses a diode or transistor in the input path, which has an exponential $I-V$ characteristic. The output voltage varies exponentially with the input voltage.

② Need / Applications :

- To expand a compressed (logarithmic) signal back to its original form.
- Used in analog computation (multiplication, division).
- Useful in signal reconstruction.
- Used in automatic gain control (AGC).
- Used in communication systems and signal processing.
- Used in waveform generation and function generators.

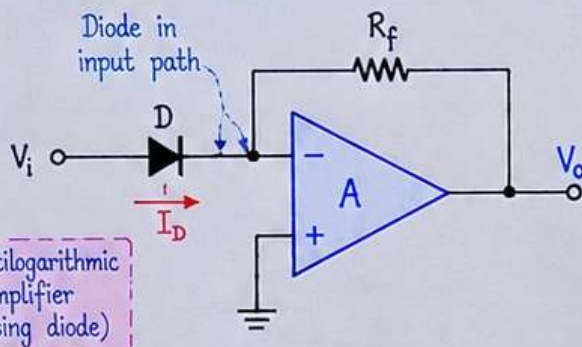
③ Basic Concept :

- A diode/transistor has an exponential current-voltage relationship :

$$I_D = I_S \left(e^{\frac{V_D}{\eta V_T}} - 1 \right)$$

- In an antilog amplifier, this exponential current is converted into a proportional output voltage using a feedback resistor R_f .
- Thus, $V_o \propto e^{V_i/\eta V_T}$.

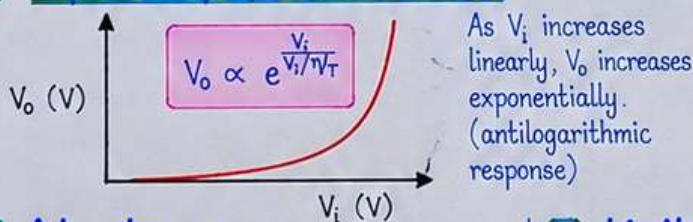
④ Circuit Diagram (Using Diode) :



⑥ Working :

- ① The input voltage V_i is applied to the diode connected at the inverting input.
- ② Due to virtual ground, the diode voltage $V_D = V_i$.
- ③ The diode current varies exponentially with V_i .
- ④ This current flows through the feedback resistor R_f , producing an output voltage $V_o = -I_D R_f$.
- ⑤ Therefore, the output voltage is proportional to the antilogarithm (exponential) of the input voltage.
- ⑥ A small change in input voltage produces a large change in output voltage (exponential response).

⑦ Input-Output Characteristic :



⑨ Advantages :

- Performs inverse of logarithmic function.
- Useful in analog computation (multiplication, division, power functions).
- Wide range of applications in signal processing.
- Simple circuit configuration.
- High sensitivity (small input change gives large output change).

⑩ Limitations :

- Accuracy depends on diode characteristics (I_S , η , V_T).
- Temperature sensitive.
- Limited input voltage range.
- Not suitable for high precision applications.
- Non-linear response.
- Affected by diode non-idealities.

⑪ Applications :

- Expansion of compressed signals (inverse of log amplifier).
- Analog computation (multiplication, division, power functions).
- Automatic gain control (AGC).
- Signal reconstruction in communication systems.
- Waveform generation.
- Audio and video signal processing.

⑫ Key Points to Remember :

- ★ Uses diode/transistor in input path.
- ★ Output is proportional to antilogarithm (exponential) of input voltage.
- ★ $V_o \approx -R_f I_S e^{V_i/\eta V_T}$ (for $V_i \gg \eta V_T$).

- ★ Inverse operation of logarithmic amplifier.
- ★ A small change in input produces a large change in output.
- ★ Widely used in analog computation and signal processing.
- ★ Temperature sensitive and limited input range.

⑤ Derivation :

- ① Current through diode :

$$I_D = I_S \left(e^{\frac{V_i}{\eta V_T}} - 1 \right)$$

- ② Since the inverting input is at virtual ground ($V \approx 0$), the diode voltage $V_D = V_i$.

- ③ This current flows through the feedback resistor R_f .
So, $V_o = -I_D R_f$

- ④ Substituting I_D ,

$$V_o = -R_f I_S \left(e^{\frac{V_i}{\eta V_T}} - 1 \right)$$

- ⑤ For $V_i \gg \eta V_T$, ($e^{V_i/\eta V_T} \gg 1$),

$$V_o \approx -R_f I_S e^{\frac{V_i}{\eta V_T}}$$

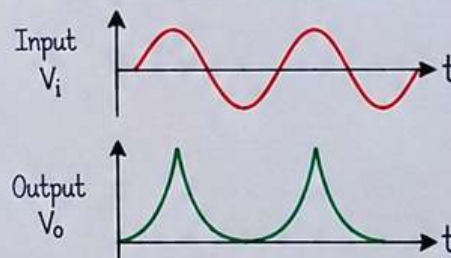
Where,

I_S = Reverse saturation current of diode

V_T = Thermal voltage (≈ 26 mV at room temperature)

η = Ideality factor (1 to 2)

⑧ Waveforms (Example) :



A sinusoidal input produces an exponentially varying output.

Astable Multivibrator (using Op-Amp)

Unit - 3
Special Purpose
Operational Amplifiers

① Definition :

An astable multivibrator is a free running comparator (oscillator) which has no stable state. It continuously switches between two quasi-stable states and generates a square wave output without any external trigger signal.

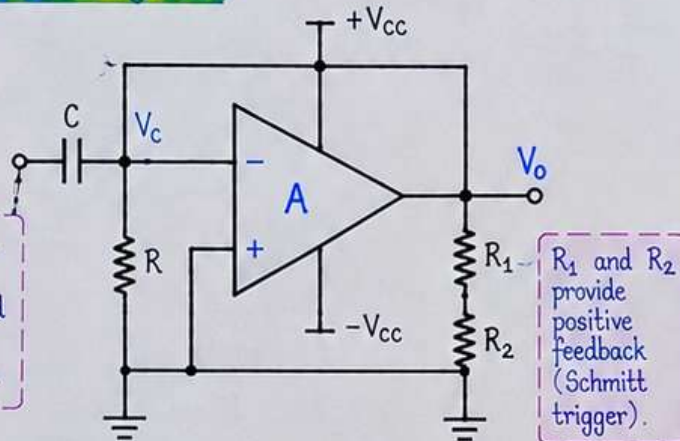
② Features :

- No stable state.
- Generates continuous square wave output.
- Frequency depends on R and C.
- Uses positive feedback (Schmitt trigger) and RC network.
- Output is alternating between $+V_{sat}$ and $-V_{sat}$.
- Simple and low cost circuit.

③ Applications :

- Square wave generator.
- Clock generator.
- Function generator.
- LED flasher.
- Pulse generation.
- Used in timers and counters.
- Wave shaping circuits.
- Relaxation oscillator.

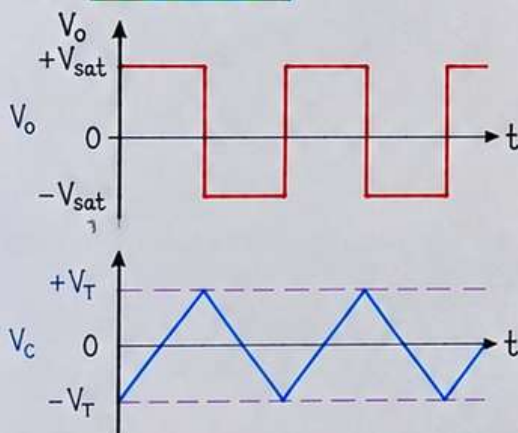
④ Circuit Diagram :



⑤ Working :

- Initially, due to positive feedback, the output saturates at $+V_{sat}$. The capacitor charges through R towards $+V_{sat}$.
- When V_c reaches the upper threshold voltage ($+V_T$), the op-amp output switches to $-V_{sat}$.
- Now the capacitor discharges through R towards $-V_{sat}$.
- When V_c reaches the lower threshold voltage ($-V_T$), the output switches back to $+V_{sat}$.
- This process repeats continuously, producing a square wave at the output and a triangular wave across the capacitor.

⑥ Waveforms :



Output voltage is a square wave and capacitor voltage is a triangular wave.

⑦ Threshold Voltages :

$$V_T = \beta V_{sat} \quad \text{where} \quad \beta = \frac{R_2}{R_1 + R_2}$$

Upper threshold = $+V_T = +\beta V_{sat}$

Lower threshold = $-V_T = -\beta V_{sat}$

⑧ Time Period and Frequency :

Time taken for charging from $-V_T$ to $+V_T$

$$T_1 = RC \ln \left(\frac{1+\beta}{1-\beta} \right)$$

Time taken for discharging from $+V_T$ to $-V_T$

$$T_2 = RC \ln \left(\frac{1+\beta}{1-\beta} \right)$$

Total time period,

$$T = T_1 + T_2 = 2RC \ln \left(\frac{1+\beta}{1-\beta} \right)$$

Frequency,

$$f = \frac{1}{T} = \frac{1}{2RC \ln \left(\frac{1+\beta}{1-\beta} \right)}$$

⑨ Derivation :

- For $+V_{sat}$, capacitor charges as

$$V_c = V_{sat} - (V_{sat} + V_T) e^{-t/RC}$$

At $V_c = +V_T$,

$$t = RC \ln \left(\frac{V_{sat} + V_T}{V_{sat} - V_T} \right)$$

$$T_1 = RC \ln \left(\frac{1+\beta}{1-\beta} \right)$$

- Similarly, for $-V_{sat}$,

$$T_2 = RC \ln \left(\frac{1+\beta}{1-\beta} \right)$$

- Therefore, $T = 2RC \ln \left(\frac{1+\beta}{1-\beta} \right)$

⑩ Advantages :

- No external trigger required.
- Simple circuit.
- Generates square wave with good stability.
- Frequency easily adjustable (by R or C).
- Useful in many applications.

⑪ Limitations :

- Output frequency not very high.
- Frequency stability depends on component values.
- Not suitable for precise timing applications.
- Sensitive to supply voltage variations.

⑫ Key Points to Remember :

- ★ Astable $\rightarrow$ no stable state (free running).
- ★ Square wave output, triangular capacitor voltage.
- ★ Threshold voltage $V_T = \beta V_{sat}$.
- ★ Time period $T = 2RC \ln \left(\frac{1+\beta}{1-\beta} \right)$.
- ★ Frequency $f = \frac{1}{2RC \ln \left(\frac{1+\beta}{1-\beta} \right)}$.
- ★ Uses positive feedback (Schmitt trigger).

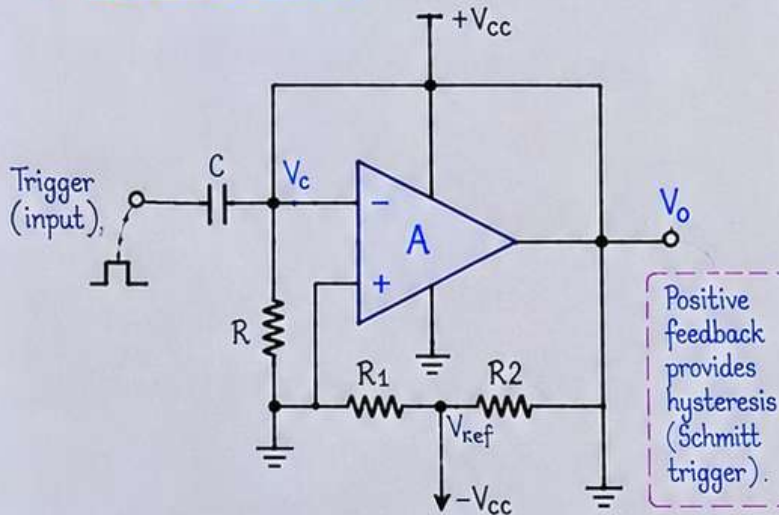
Monostable Multivibrator (using Op-Amp)

Unit - 3
Special Purpose
Operational Amplifiers

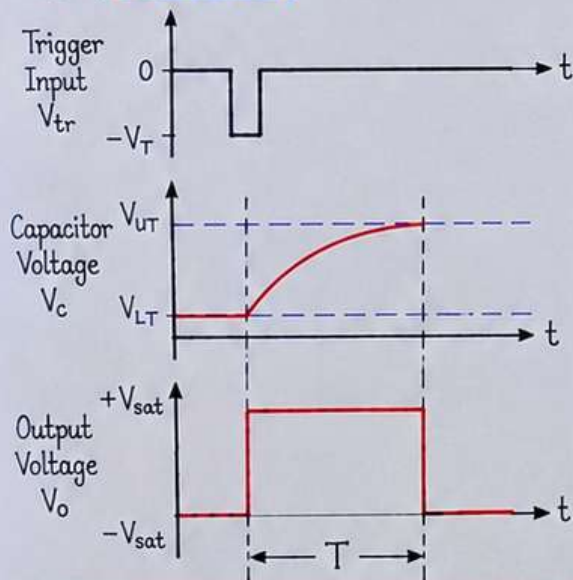
① Definition :

A monostable multivibrator is a circuit which has one stable state and one quasi-stable state. It remains in the stable state indefinitely and when an external trigger is applied, it goes to the quasi-stable state for a definite time period and then automatically returns to the stable state.

④ Circuit Diagram :



⑥ Waveforms :



A negative trigger produces a positive output pulse (for the shown configuration).

② Features :

- One stable state and one quasi-stable state.
- Requires an external trigger.
- Generates a single output pulse of fixed width.
- Used for time delay and pulse generation.
- The time period is determined by R and C.
- Simple and low cost circuit.

⑤ Working :

- Initially the circuit is in stable state. Output is at $+V_{sat}$ (or $-V_{sat}$ depending on design).
- When a negative trigger pulse is applied, the capacitor C rapidly discharges, making V_c cross the lower threshold voltage (V_{LT}).
- The op-amp output switches to the opposite saturation level ($-V_{sat}$).
- Now the capacitor charges through R towards $+V_{sat}$ (or $-V_{sat}$).
- When V_c reaches the upper threshold voltage (V_{UT}), the output switches back to its original state.
- The output returns to the stable state and remains there until next trigger.

⑦ Threshold Voltages :

$$V_{UT} = \beta V_{sat} \quad V_{LT} = -\beta V_{sat}$$

$$\text{where, } \beta = \frac{R_2}{R_1 + R_2}$$

⑧ Time Period (Pulse Width) :

The time for which the output remains in the quasi-stable state.

$$T = RC \ln \left(\frac{1+\beta}{1-\beta} \right)$$

$$\text{where, } \beta = \frac{R_2}{R_1 + R_2}$$

⑨ Derivation :

During charging of capacitor,
 $V_c = V_{sat} - (V_{sat} - V_{LT}) e^{-t/RC}$
 At $t = T$, $V_c = V_{UT}$
 $V_{UT} = V_{sat} - (V_{sat} - V_{LT}) e^{-T/RC}$
 Solving, $T = RC \ln \left(\frac{1+\beta}{1-\beta} \right)$

⑩ Example :

Given $R = 100 \text{ k}\Omega$, $C = 0.1 \text{ }\mu\text{F}$
 $R_1 = 10 \text{ k}\Omega$, $R_2 = 90 \text{ k}\Omega$
 $\beta = \frac{90}{10+90} = 0.9$
 $T = (100 \times 10^3)(0.1 \times 10^{-6}) \times \ln \left(\frac{1+0.9}{1-0.9} \right)$
 $T = 0.69 \text{ ms}$

⑪ Advantages :

- Generates accurate fixed-width pulses.
- Simple and low cost.
- Good noise immunity (due to Schmitt trigger).
- Wide range of applications.
- Can be easily designed using R and C.

⑫ Limitations :

- Pulse width depends on component tolerances.
- Not suitable for very high frequency applications.
- Requires external trigger.
- Output pulse width is fixed (unless R or C is changed).
- Sensitive to supply voltage variations.

⑬ Key Points to Remember :

- ★ One stable state, one quasi-stable state.
- ★ Requires an external trigger.
- ★ Generates a single pulse of fixed width.
- ★ $T = RC \ln \left(\frac{1+\beta}{1-\beta} \right)$
- ★ Threshold voltages : $\pm \beta V_{sat}$
- ★ Used for time delay and pulse generation.

Bistable Multivibrator (using Op-Amp)

Unit - 3
Special Purpose
Operational Amplifiers

① Definition :

A bistable multivibrator is a switching circuit which has two stable states. It remains in either of the two stable states ($+V_{sat}$ or $-V_{sat}$) indefinitely until an external trigger signal is applied to change the state. It is also called a flip-flop or a Schmitt trigger used as a memory element.

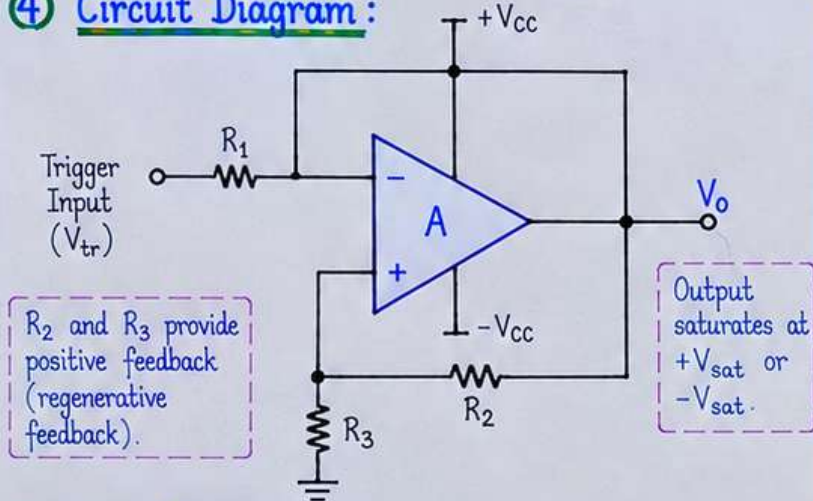
② Features :

- Two stable states.
- Requires external trigger to change the state.
- Acts as a memory element (stores 1 bit of information).
- Uses positive feedback (regenerative feedback).
- Simple circuit and low cost.
- Wide range of applications.

③ Applications :

- Memory circuits (stores binary data).
- Switching circuits.
- Wave shaping.
- Debouncing switches.
- Digital logic circuits.
- Comparators with hysteresis (Schmitt trigger).
- Used in counters, registers.
- Used in on-off control systems.

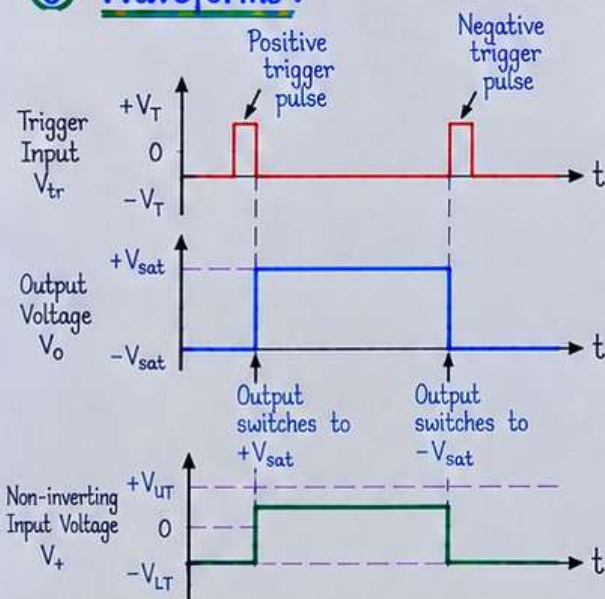
④ Circuit Diagram :



⑤ Working :

- ① Due to positive feedback, the op-amp has two stable output states: $+V_{sat}$ and $-V_{sat}$.
- ② The non-inverting input voltage (V_+) is a fraction of the output voltage due to the potential divider (R_2, R_3).
- ③ When a positive trigger pulse is applied ($V_{tr} > V_{UT}$), the output switches to $+V_{sat}$.
- ④ When a negative trigger pulse is applied ($V_{tr} < V_{LT}$), the output switches to $-V_{sat}$.
- ⑤ After the trigger is removed, the output remains in the new state (no automatic change).
- ⑥ Thus, the circuit acts as a bistable multivibrator with two stable states.

⑥ Waveforms :



A positive trigger makes the output $+V_{sat}$ and a negative trigger makes the output $-V_{sat}$.

⑦ Threshold Voltages :

The non-inverting input voltage V_+ is given by

$$V_+ = \frac{R_3}{R_2 + R_3} V_o$$

When output is at $+V_{sat}$,

$$V_{UT} = \frac{R_3}{R_2 + R_3} (+V_{sat})$$

When output is at $-V_{sat}$,

$$V_{LT} = \frac{R_3}{R_2 + R_3} (-V_{sat})$$

Hysteresis width (V_H) :

$$V_H = V_{UT} - V_{LT} = \frac{2 R_3}{R_2 + R_3} V_{sat}$$

⑧ Switching Condition :

- If $V_{tr} > V_{UT} \Rightarrow$ Output = $+V_{sat}$ (positive trigger).
- If $V_{tr} < V_{LT} \Rightarrow$ Output = $-V_{sat}$ (negative trigger).
- If $V_{LT} < V_{tr} < V_{UT} \Rightarrow$ Output remains in previous state (no change).

⑨ Example :

Given: $V_{sat} = 12 \text{ V}$, $R_2 = 30 \text{ k}\Omega$, $R_3 = 10 \text{ k}\Omega$.

Find V_{UT} , V_{LT} and V_H .

Solution :

$$V_{UT} = \frac{10}{30 + 10} \times 12 = 3 \text{ V}$$

$$V_{LT} = -3 \text{ V}$$

$$V_H = 6 \text{ V}$$

⑩ Advantages :

- Two stable states (acts as memory).
- Simple and low cost.
- Good noise immunity (due to hysteresis).
- Wide range of applications.
- Can be easily designed using resistors.

⑪ Limitations :

- Requires external trigger signal to change the state.
- Not suitable for generating continuous waveforms.
- Switching speed is limited by the op-amp and circuit components.
- Output is limited to saturation levels.

⑫ Key Points to Remember :

- ★ Bistable $\rightarrow$ two stable states.
- ★ Uses positive feedback.
- ★ Output remains in the new state after trigger is removed.
- ★ Threshold voltages: $V_{UT} = \frac{R_3}{R_2 + R_3} (+V_{sat})$
 $V_{LT} = \frac{R_3}{R_2 + R_3} (-V_{sat})$
- ★ Hysteresis width: $V_H = \frac{2 R_3}{R_2 + R_3} V_{sat}$
- ★ Used as a memory element (flip-flop).

Comparison + Important Formula

Unit - 3
Special Purpose
Operational Amplifiers

(Op-Amp Based Circuits)

① Comparison of Multivibrators (Op-Amp)

Parameter	Astable Multivibrator	Monostable Multivibrator	Bistable Multivibrator
No. of stable states	No stable state (free running)	One stable state, one quasi-stable state	Two stable states
Trigger required	Not required	Required (external trigger)	Required (to change state)
Output waveform	Continuous square wave	Single pulse of fixed width	Two-level output (+V _{sat} or -V _{sat})
Time period / Pulse width	$T = 2RC \ln \left(\frac{1+\beta}{1-\beta} \right)$	$T = RC \ln \left(\frac{1+\beta}{1-\beta} \right)$	No time period (maintains state until next trigger)
Frequency	$f = \frac{1}{T}$	$f = \frac{1}{T}$ (for triggered pulses)	Not applicable
Main components	R, C (RC network)	R, C + external trigger	R ₁ , R ₂ , R ₃ (positive feedback)
Feedback type	Positive feedback (Schmitt trigger)	Positive feedback (Schmitt trigger)	Positive feedback (Schmitt trigger)
Applications	Clock generator, waveform generator, LED flasher	Pulse generator, time delay circuits, timers	Memory circuits, switching circuits, comparators with hysteresis

② Important Circuits, Waveforms and Formulas

Astable Multivibrator

Output Waveform :

Time Period :

$$T = 2RC \ln \left(\frac{1+\beta}{1-\beta} \right)$$

Frequency :

$$f = \frac{1}{2RC} \left(\frac{1+\beta}{1-\beta} \right)$$

$$\beta = \frac{R_2}{R_1 + R_2}$$

→ Generates continuous square wave.

Monostable Multivibrator

Output Waveform :

Pulse Width :

$$T = RC \ln \left(\frac{1+\beta}{1-\beta} \right)$$

$$\beta = \frac{R_2}{R_1 + R_2}$$

♦ Generates a single pulse of fixed width (on triggering).

Bistable Multivibrator

Output Waveform :

Threshold Voltages :

$$V_{UT} = \frac{R_3}{R_2 + R_3} (+V_{sat})$$

$$V_{LT} = \frac{R_3}{R_2 + R_3} (-V_{sat})$$

Hysteresis Width :

$$V_H = V_{UT} - V_{LT} = \frac{2R_3}{R_2 + R_3} V_{sat}$$

→ Acts as a memory element (two stable states).

③ Important Formulas (Summary Box)

- Time period (Astable) : $T = 2RC \ln \left(\frac{1+\beta}{1-\beta} \right)$
- Pulse width (Monostable) : $T = RC \ln \left(\frac{1+\beta}{1-\beta} \right)$
- Frequency : $f = \frac{1}{2RC} \left(\frac{1+\beta}{1-\beta} \right)$
- Feedback factor : $\beta = \frac{R_2}{R_1 + R_2}$ (Astable/Monostable)
- Threshold voltages (Bistable) : $V_{UT} = \frac{R_3}{R_2 + R_3} (+V_{sat})$
 $V_{LT} = \frac{R_3}{R_2 + R_3} (-V_{sat})$
- Hysteresis width : $V_H = \frac{2R_3}{R_2 + R_3} V_{sat}$

④ Other Important Formulas (from Previous Parts)

- Instrumentation Amplifier (3-op-amp) : $V_{out} = \left(1 + \frac{2R}{R_G} \right) (V_2 - V_1)$
- V-I Converter : $I_o = -\frac{V_i}{R}$
- I-V Converter : $V_o = -I_f R_f$
- Logarithmic Amplifier : $V_o = -\eta V_T \ln \left(\frac{V_i}{R I_s} \right)$
- Antilogarithmic Amplifier : $V_o = -R_f I_s e^{\frac{V_i}{\eta V_T}}$
- Schmitt Trigger (Threshold) : $V_T = \pm \beta V_{sat}$ ($\beta = \frac{R_2}{R_1 + R_2}$)

⑤ Key Takeaways

- ★ Astable → continuous square wave (no trigger).
- ★ Monostable → single pulse of fixed width (requires trigger).
- ★ Bistable → two stable states (memory element).
- ★ Formulas depend on circuit configuration and resistor ratios.
- ★ These circuits use positive feedback (Schmitt trigger).

Practice
Diagrams
+
Formulas
=
Better
Understanding
😊

"Electronics
Builds
tomorrow."

Unit - 3 (Operational Amplifiers)

Diploma-Level Important Questions + Numericals

Unit - 3
Special Purpose
Operational Amplifiers

(A) Important Theory Questions

- Define operational amplifier. Draw its symbol and mention ideal characteristics.
- What is a Schmitt trigger? Explain with circuit.
- Explain the need of instrumentation amplifier.
- Draw and explain the 3-op-amp instrumentation amplifier. Derive its gain.
- What is V-I converter? Draw its circuit and explain working.
- What is I-V converter? Draw its circuit and explain working.
- Explain logarithmic amplifier with neat diagram.
- Explain antilogarithmic amplifier with neat diagram.
- What is an astable multivibrator? Explain with circuit and waveforms.
- What is a monostable multivibrator? Explain with circuit and waveforms.
- What is a bistable multivibrator? Explain with circuit and waveforms.
- Compare astable, monostable and bistable multivibrators.
- What is a precision rectifier? Explain its working.
- What is a peak detector? Explain with circuit.
- Explain sample and hold circuit.
- What is a window comparator? Draw and explain.
- Explain the use of op-amp in timer and waveform generation circuits.
- List the applications of special purpose op-amp circuits.
- What are the limitations of op-amp based circuits?
- Draw neat waveforms for all multivibrators and label them properly.

(C) Important Circuit-Based Questions

- Draw the circuit of a 3-op-amp instrumentation amplifier and derive its gain.
- Draw the circuit of V-I and I-V converters.
- Draw the circuit and waveforms of logarithmic and antilogarithmic amplifiers.
- Draw the circuit of an astable multivibrator and derive its time period.
- Draw the circuit of a monostable multivibrator and derive the pulse width.
- Draw the circuit of a bistable multivibrator and find the threshold voltages.
- Draw the circuit of a Schmitt trigger and explain its working.
- Draw the circuit of a precision rectifier.
- Draw the circuit of a peak detector using op-amp.
- Draw the circuit of a sample and hold circuit.

(D) Tips to Solve Numericals

- Identify the type of circuit and write the basic formula.
- Convert all values to proper units (Ω , F, V, A).
- Use $\ln 2 = 0.693$ where required.
- For log/antilog problems, use $V_T = 26 \text{ mV}$ (at room temperature).
- Draw the circuit and label all values.
- Check the sign of output voltage (inverting / non-inverting).
- Write final answer with proper unit.

(B) Numericals (with Answers)

- Find the gain of an instrumentation amplifier if $R_1 = 10 \text{ k}\Omega$ and $R_g = 1 \text{ k}\Omega$.
Sol: $G = 1 + \frac{2R_1}{R_g} = 1 + \frac{2(10\text{k})}{1\text{k}} = 1 + 20 = 21$
Gain = 21
- In a V-I converter, $V_{in} = 5 \text{ V}$ and $R = 2 \text{ k}\Omega$. Find the output current I_o .
Sol: $I_o = \frac{V_{in}}{R} = \frac{5}{2000} = 2.5 \text{ mA}$
 $I_o = 2.5 \text{ mA}$
- In an I-V converter, input current is $100 \mu\text{A}$ and feedback resistor $R_f = 10 \text{ k}\Omega$. Find output voltage.
Sol: $V_o = -I_f R_f = -(100 \times 10^{-6})(10 \times 10^3) = -1 \text{ V}$
 $V_o = -1 \text{ V}$
- For a logarithmic amplifier, input voltage $V_i = 0.2 \text{ V}$, $V_T = 26 \text{ mV}$. Find output voltage.
Sol: $V_o = -\eta V_T \ln \left(\frac{V_i}{V_d} \right)$
Assume $\eta = 1$, $V_d = 26 \text{ mV}$
 $V_o = -(26 \text{ mV}) \ln \left(\frac{0.2}{0.026} \right) = -26 \text{ mV} \ln (7.692) = -26 \text{ mV} \times 2.04 \approx -53 \text{ mV}$
 $V_o \approx -53 \text{ mV}$
- For an astable multivibrator, $R_2 = 10 \text{ k}\Omega$, $C = 0.1 \mu\text{F}$. Find time period and frequency.
Sol: $T = 2RC \ln \left(\frac{1+\beta}{1-\beta} \right)$, for $\beta = \frac{1}{3}$
 $T = 2(10\text{k})(0.1\mu\text{F}) \ln 2 = 2 \times 10^4 \times 10^{-7} \times 0.693 = 1.386 \text{ ms}$
 $f = \frac{1}{T} = \frac{1}{1.386 \text{ ms}} = 721 \text{ Hz}$
 $T = 1.386 \text{ ms}$, $f \approx 721 \text{ Hz}$
- For a monostable multivibrator, $R_2 = 100 \text{ k}\Omega$, $C = 0.01 \mu\text{F}$. Find the pulse width (T).
Sol: $T = RC \ln \left(\frac{1+\beta}{1-\beta} \right)$, for $\beta = \frac{1}{3}$
 $T = (100\text{k})(0.01\mu\text{F}) \ln 2 = 10^{-3} \times 0.693 = 0.693 \text{ ms}$
 $T = 0.693 \text{ ms}$
- For a bistable multivibrator, $R_2 = 20 \text{ k}\Omega$, $R_3 = 10 \text{ k}\Omega$, $V_{sat} = 12 \text{ V}$. Find threshold voltages.
Sol: $V_{UT} = \frac{R_3}{R_2 + R_3} V_{sat} = \frac{10}{30} \times 12 = 4 \text{ V}$
 $V_{LT} = -4 \text{ V}$
 $V_{UT} = +4 \text{ V}$, $V_{LT} = -4 \text{ V}$

(E) Key Formulas (Revision)

- Instrumentation Amplifier :
 $G = 1 + \frac{2R_1}{R_g}$
- V-I Converter :
 $I_o = \frac{V_{in}}{R}$
- I-V Converter :
 $V_o = -I_f R_f$
- Logarithmic Amplifier :
 $V_o = -\eta V_T \ln \left(\frac{V_i}{V_d} \right)$
- Antilogarithmic Amplifier :
 $V_o = -R_f I_s e^{\frac{V_i}{\eta V_T}}$
- Astable Multivibrator :
 $T = 2RC \ln \left(\frac{1+\beta}{1-\beta} \right)$ $f = \frac{1}{T}$
- Monostable Multivibrator :
 $T = RC \ln \left(\frac{1+\beta}{1-\beta} \right)$
- Bistable Multivibrator :
 $V_{UT} = \frac{R_3}{R_2 + R_3} (+V_{sat})$
 $V_{LT} = \frac{R_3}{R_2 + R_3} (-V_{sat})$
 $V_H = \frac{2R_3}{R_2 + R_3} V_{sat}$
- Schmitt Trigger :
 $V_T = \pm \beta V_{sat}$ $\left(\beta = \frac{R_2}{R_1 + R_2} \right)$

(F) Expected Exam Questions

- ★ Derive the gain of 3-op-amp instrumentation amplifier.
- ★ Explain the working of V-I and I-V converters with neat diagrams.
- ★ Derive the expression for output voltage of logarithmic and antilogarithmic amplifiers.
- ★ Analyze the op-amp as astable, monostable and bistable multivibrator with waveforms.
- ★ Compare different multivibrators.
- ★ Numerical on time period, pulse width, threshold voltage and gain.
- ★ Explain applications of special purpose op-amp circuits.
- ★ Draw neat and labeled waveforms for all circuits.
- ★ Short notes on Schmitt trigger, precision rectifier, peak detector and sample & hold.

Practice Numericals Daily
for Better Understanding
and Higher Marks!

